BTS–VII(S)–04.19–0708 Reg. No.

A

# B.Tech. Degree VII Semester Supplementary Examination April 2019

## IT 15-1703 COMPUTER GRAPHICS

*(2015 Scheme)*

Time : 3 Hours Maximum Marks : 60

### PART A

(Answer *ALL* questions)

(10 × 2 = 20)

I. (a) Draw a line with end points (12,8) and (20,14) using Bresenham's line drawing algorithm.

(b) Draw a line with end points (0,0) and (8,4) using DDA algorithm.

(c) Explain 2D Scaling with necessary equations and examples.

(d) Consider a rectangle with vertices A(50,100), B(100,100), C(100,200) and D(50,200). Apply translation on the above rectangle with translation distances 12 and 14 about x and y directions respectively and find the new vertices after translation.

(e) Discuss briefly about Bezier curves.

(f) Differentiate between Approximation and Interpolation Splines.

(g) Discuss briefly about various types of polygon tables using for polygon surfaces representation with examples.

(h) If a pyramid with vertices A(-0.5,0,0.5), B(0.5,0,0.5), C(0.5,0,-0.5), D(-0.5,0,-0.5) and E(0,1,0) is scaled by values 3,1, and 5 in x,y and z directions respectively, find the new vertices after scaling along with necessary equations.

(i) Discuss briefly about Gouraud Shading.

(j) Discuss briefly any two 3D display methods.



### PART B

(4 × 10 = 40)

II. Write Midpoint circle algorithm. Trace the algorithm to draw a circle with radius 30 and centre (20,30). (10)

**OR**

III. Draw the functional block diagram of CRT and explain its different parts. (10)

IV. (a) Discuss window to viewport transformation. (5)

(b) Explain Raster Methods for Transformation. (5)

**OR**

V. Illustrate Sutherland Hodgeman Polygon Clipping Algorithm with an example. (10)

VI. (a) Discuss Constructive Solid geometry methods in detail. (5)

(b) Explain Sweep Representation. (5)

**OR**

VII. What are Fractals? Explain different types of fractals and their construction methods. (10)

VIII. Differentiate between Z Buffer Method and A Buffer Method in detail. (10)

**OR**

IX. (a) Discuss in detail about Scan Line Method for visible surface Detection with necessary diagrams. (5)

(b) Discuss about BSP Tree method for visible surface detection with an example. (5)

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